

Samy Haffoudhi

PhD Student in Natural Language Processing

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Research interests

Information extraction and entity linking · Knowledge graphs · Reasoning and generalization in LLMs · LLM post-training (RL, distillation) · Mechanistic interpretability

Publications

- Submitted **When Does Reasoning Help Entity Disambiguation? Small LLMs In-Domain and Under Shift**
S. Haffoudhi, F. M. Suchanek, N. Holzenberger. Under review, 2026.
- ISWC 2026 **LELA: an LLM-based Entity Linking Approach with Zero-Shot Domain Adaptation**
S. Haffoudhi, F. M. Suchanek, N. Holzenberger. *International Semantic Web Conference (ISWC)*, Bari, Italy, 2026 (to appear). arXiv:2601.05192
- IJCAI 2026 **LELA: An End-to-end LLM-based Entity Linking Framework with Zero-shot Domain Adaptation**
S. Haffoudhi, N. Dobričić, F. M. Suchanek, N. Holzenberger. *Int. Joint Conference on Artificial Intelligence (IJCAI-ECAI)*, Demo Track, Bremen, Germany, 2026. doi:10.24963/ijcai.2026/966

Experience

- Sep. – Nov. 2026 **Visiting Researcher**, *Sapienza University of Rome*, Sapienza NLP group, Rome, Italy.
3-month research visit hosted by Roberto Navigli.
- Sep. 2025 – Feb. 2028 **Doctoral Researcher**, *Télécom Paris, Institut Polytechnique de Paris*, DIG team, Plateau de Saclay, France.
• Project: Knowledge-based Language Models; advisors: Fabian M. Suchanek and Nils Holzenberger (expected)
• Designed LELA, a modular LLM-based entity linking method with zero-shot domain adaptation ([code](#))
- Jan. 2023 – Jun. 2024 **Student Software Engineer (part-time)**, *Microsoft*, Lyngby, Denmark.
Developer Tools team of Microsoft Dynamics 365 Business Central: AL language compiler and Visual Studio Code extension. Large C# and TypeScript codebase, test-driven development, bug fixes and new features.
- Apr. – Sep. 2022 **Research Intern**, *Thales*, Plateau de Saclay, France.
Deep metric learning and multitask learning for audio: voice activity detection, audio signal classification and source separation.

Teaching and service

- 2025–2026 TA for 10 courses, incl. *Language Modeling* and *Large-scale Generative Models*; project supervision.
- Service Reviewer for AKBC 2026 (Automated Knowledge Base Construction) and *Machine Learning* (Springer).

Education

- 2025–2028 **PhD in Computer Science**, *Télécom Paris, Institut Polytechnique de Paris*, expected graduation: Feb. 2028.
- 2022–2024 **MSc in Computer Science and Engineering**, *DTU (Technical University of Denmark)*, Lyngby, Denmark.
Thesis: *Implementation and evaluation of recent neuroevolution algorithms*, supervised by Carsten Witt.
- 2020–2024 **MEng (Diplôme d'ingénieur)**, *École Centrale de Nantes*, Nantes, France.
A highly selective French School of Engineering. **Data analysis and processing specialization:** statistics, machine learning, signal and image processing, algorithms, databases.
- 2018–2020 **“Classes préparatoires”** (mathematics, computer science, physics), *Lycée Janson de Sailly*, Paris, France.

Skills and languages

- Programming Python, Rust, C/C++, C#, Java, OCaml, Bash, SQL
- ML and tools Hugging Face Transformers, vLLM, TransformerLens, Linux, Git, Docker, $\text{\LaTeX}$
- Languages **French** (native), **English** (fluent, TOEFL 2022: 108/120), German, Spanish and Arabic (basics)